

UAV-X: Resilient BVLOS Swarm Challenge

Stage 1 — Preliminary Design Verification: Technical Proposal

Team ID	[Enter team ID]
Team Name	[Enter team name]
Team ID / Registration No.	[As issued on techfest.org]
Institution / Organization	[College / University / Startup]
Team Leader	[Name · email · phone]
Members (up to 5)	[Name 1, Name 2, Name 3, Name 4, Name 5]
Category	[UG / PG / Research scholar / Startup]
Submission Date	[On or before 27 September 2026]
Track	GC-1 · UAV-X: Resilient BVLOS Swarm Challenge

Contents

1. Abstract / Executive Summary

Half a page. Written last, read first.

Summarise the mission problem, your core technical approach to resilient BVLOS swarm operation, and your headline proof-of-concept results (e.g. mission completion rate, packet delivery ratio, recovery time). State clearly what is novel in your solution.

[Your abstract here — approx. 150–250 words.]

2. Problem Understanding & Mission Analysis

Show that you understand the disaster-response scenario and its constraints. This grounds the rest of the report.

2.1 Mission scenario and assumptions

Restate the mission in your own words: GCS outside the affected area, a fleet of limited-endurance UAVs surveying Points of Interest (PoIs), collecting imagery/situational data, and relaying it to the GCS over a resilient multi-hop aerial link. List the assumptions

you make about map size, number of UAVs, endurance, comms range, and PoI distribution.

[...]

2.2 Requirements and objectives

Translate the mission into concrete, testable requirements the swarm must satisfy:

- Survey all assigned disaster locations (PoIs).
- Maintain end-to-end connectivity with the GCS.
- Dynamically assign relay UAVs.
- Reconfigure the network when links degrade, UAVs fail, or UAVs return home to recharge.
- Prioritise newly emerging high-priority regions.
- Complete the mission safely, within the allotted time, without leaving the geofence or running out of charge.

2.3 Key challenges and trade-offs

Identify the hard problems (e.g. coverage-vs-connectivity trade-off, endurance-vs-relay allocation, latency under multi-hop) and how you intend to address them.

[...]

3. System & Software Architecture

Required deliverable — include a labelled architecture diagram.

3.1 System overview

Provide a block diagram of the overall system: GCS, UAV agents, sensing payload, communication layer, and the autonomy stack. Caption it as Figure 1.

[Insert Figure 1: System architecture]

3.2 Software architecture

Describe your software stack and module decomposition — e.g. ROS/ROS 2 nodes, planners, estimators, comms manager, mission manager, GCS interface. State the middleware, languages (Python/C++), and key libraries. Show data flow and the interfaces between modules.

[Insert Figure 2: Software module/data-flow diagram]

3.3 Autonomy pipeline

Explain the decision loop each UAV runs: perception → situational awareness → task/role assignment → planning → control, and how the GCS coordinates (centralised, decentralised,

or hybrid). Justify your choice.

[...]

4. Swarm Autonomy & Mission Planning

Describe how the swarm surveys all PoIs efficiently within endurance limits. Cover task allocation across UAVs, coverage/path planning, revisits, and how high-priority regions that emerge mid-mission are prioritised and re-planned.

4.1 Task allocation and coverage

[Algorithm/approach, with a short justification.]

4.2 Dynamic re-tasking and priority handling

[How new high-priority PoIs are inserted and the plan updated.]

4.3 Endurance and recharging management

Explain how you track battery/endurance, when a UAV returns to home to recharge, and how its tasks are handed off so the mission continues.

[...]

5. Resilient BVLOS Communication

This is the heart of the challenge. Describe how you maintain an end-to-end, multi-hop aerial link between every operating UAV and the GCS, and how the network stays connected as UAVs move, fail, or leave to recharge.

5.1 Communication model and assumptions

State your link/propagation model, range, bandwidth, and the standardised communication assumptions provided by the organizers that you rely on.

[...]

5.2 Multi-hop relay network design

Explain the network topology, routing, and how connectivity is measured and maintained (e.g. connectivity availability, packet delivery ratio, latency budgets).

[...]

5.3 Connectivity-aware planning

Describe how communication constraints shape motion planning — i.e. how coverage decisions account for the need to stay connected to the GCS.

[...]

6. Autonomous Relay & Role Management

Explain how UAVs autonomously switch between roles (surveyor vs. relay), how relays are assigned and reassigned as the swarm moves, and how role decisions are made (centralised at GCS, decentralised, or hybrid). Report expected relay-reallocation behaviour.

[...]

7. Safety & Collision Avoidance

Maps to Safety & Collision Avoidance — 10% of the score.

Describe inter-UAV collision avoidance, minimum-separation guarantees, geofence enforcement, and the charge-safety rule (no UAV should be left without charge). Report the safety mechanisms and the minimum inter-UAV separation you maintain.

[...]

8. Innovation & Technical Merit

Maps to Innovation & Technical Merit — 5% of the score.

State clearly what is novel or technically distinctive about your approach relative to standard baselines, and why it matters for resilient BVLOS swarm operation. Cite prior work you build on or improve upon.

[...]

9. Simulation Framework & Proof-of-Concept

Required deliverable — a working proof-of-concept simulation.

Name the open-source framework(s) you use — e.g. PX4 SITL, ArduPilot SITL, Gazebo, AirSim, Isaac Sim, ROS/ROS 2, Webots, CoppeliaSim, ns-3, or equivalent — and describe the proof-of-concept scenario you ran. Include screenshots or plots and summarise what the demonstration video shows.

9.1 Simulation setup

Frameworks, number of UAVs, map/scenario, communication model, and how it maps to the organizers' benchmark scenarios and standardised log format.

[...]

9.2 Proof-of-concept results

[Insert Figure 3: PoC results (trajectories, connectivity, coverage over time).]

10. Evaluation & Performance Metrics

Report your proof-of-concept results against the official performance metrics. Use the organizers' standardised log format. State the scenario each number was measured on.

Category	Performance metric	Your result
Mission	Completion rate	[%]
Mission	Completion time	[s]
Mission	Priority-weighted mission score	[...]
Communication	Packet delivery ratio	[%]
Communication	Latency	[ms]
Communication	Connectivity availability	[%]
Communication	Communication downtime	[s]
Autonomy	Relay reallocations	[count]
Autonomy	Recovery time	[s]
Autonomy	Network reconfiguration efficiency	[...]
Robustness	Performance after UAV / link failures	[...]
Safety	Collision count	[count]
Safety	Minimum inter-UAV separation	[m]

Also confirm the hard constraints: no UAV ran out of charge, and no UAV left the geofenced area.

11. Reproducibility

Required deliverables — source code, installation instructions, demonstration video.

11.1 Source code

State the repository/archive you are submitting, its structure, licence, and any third-party dependencies. Ensure the submitted code reproduces the results reported above.

[Repository name / archive; brief structure.]

11.2 Installation & run instructions

Give exact, ordered steps to set up the environment and run the proof-of-concept, so an evaluator can reproduce it. Include OS, framework versions, and commands.

- [Step 1 — environment setup]
- [Step 2 — build / install]
- [Step 3 — run the demo scenario]

11.3 Demonstration video

Provide the link to your demonstration video and a one-line description of what it shows. Ensure the link is accessible to the evaluators.

[Video link]

12. Team & Contributions

List each member and their primary contribution (controls, networking, planning, AI/optimization, integration, etc.). Interdisciplinary contributions are encouraged.

[Member — role/contribution]

13. References

Cite prior work, tools, and datasets you build on. Use a consistent style (IEEE recommended). Number references and cite them in the text as [1], [2], ...

[1] [...]

[2] [...]

Stage 1 Submission Checklist

Confirm every item before submitting by email to pushpak_gc2026@aero.iitb.ac.in on or before 27 September 2026.

	Deliverable	Notes
☐	Technical proposal (this document)	6–8 pages, PDF
☐	Software architecture	Diagram + description (in §3 or separate)
☐	Working proof-of-concept simulation	Runs on a standard open-source framework
☐	Source code	Reproduces reported results

☐	Installation instructions	Exact, ordered steps
☐	Demonstration video	Accessible link